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SoilSense: An IoT-Based Soil Health Monitoring System for Smallholder Farmers in Bangladesh to Support Sustainable Agriculture

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***Abstract*—Smallholder farmers have limited access to soil diagnostics and scientific advice in a timely fashion, which results in ineffective use of fertilizers, deteriorating soil health and decreasing crop yields. In a bid to solve these problems, this paper proposes SoilSense, a low-cost, IoT-enabled soil health monitoring system to assist precision agriculture in resource-limited environments. The system consists of an ESP32 microcontroller with DHT11 sensor (to measure temperature and humidity), an analog rain sensor, and a capacitive soil moisture sensor that measure real-time environmental parameters. Readings are shown on an I2C LCD and sent to a Telegram bot which helps to monitor in real-time as well as to keep track of long-term logging. The collected data is processed by a machine learning model developed using regression in order to predict the necessary levels of soil nutrients Nitrogen (N), Phosphorus (P) and Potassium (K). Based on these predictions, the system can produce localized soil improvement advice in Bengali based on the 'Union Sahayika' agricultural recommendations. SoilSense provides a scalable and easy to use solution to enhance soil management and decision-making to smallholder farmers, aligning with SDG 2 (Zero Hunger) and SDG 15 (Life on Land) to achieve sustainable agriculture.**

***Index Terms*—IoT-based-soil-monitoring, Sustainable-farming, Soil-nutrient-prediction, Telegram-bot-integration, NPK-estimation, Smallholder-farmer-support**

I. Introduction

Healthy soil is not just a scientific notion to a lot of smallholder farmers, but rather a daily challenge that affects their livelihoods, families, and communities. Poor availability of feasible farming advice and use of traditional farming systems usually results in declining soil health, nutrient disorders and reductions in yield. Such issues drive the farmers to tight corners, leaving them with another option in some cases, which is to encroach farming into the natural lands, causing a negative effect on biodiversity and ecological stability. It is with the realization of these inter-linked challenges that easily understandable, readily available tools are needed to enable farmers make informed sustainable agricultural decisions. It is based on this vision that we present SoilSense, an IoT enabled soil health monitoring system, specifically conceived to help farmers, with limited resources, in real-time. By using inexpensive, easily obtainable technology an ESP32 microcontroller with basic temperature, humidity, rain, and soil moisture sensors SoilSense can constantly and dependably deliver current information on soil conditions. In addition, the system will incorporate state-of-the-art machine learning models to provide precise forecasting of essential soil nutrients Nitrogen, Phosphorus, Potassium. SoilSense overcomes this barrier to complex agricultural science by simply relaying personalized advice in Bengali through a Telegram messaging interface. Finally, besides assisting farmers to raise productivity and to better manage their soils, this solution actively contributes to the wider agenda of environmental conservations, biodiversity protection and sustainable growth of agricultural activities.

II. Literature Review

A. IoT-Based Soil Monitoring Systems

Real-time soil monitoring and precision management have become a reality in agriculture due to the incorporation of the Internet of Things (IoT) technology in agriculture. Naik et al. presented a framework of real-time environmental testing, and defined the significance of the constant monitoring of the soil health using the embedded sensor networks, and wireless communication modules [1]. Some of them are based on micro controllers, such as Arduino or ESP32 as a central node to aggregate sensor data and allow remote access [2], [3], [4]. Indeed, research activities demonstrated that IoT-enabled soil and water testing systems can increase the precision and cadence of the field measurements that are beyond the capability of the conventional manual sampling and laboratory testing [5], [6]. The solutions usually encompass cloud computing and easy-to-operate interfaces to visualize the results in real-time, helping to make smarter and more timely farming decisions [7], [8].

B. Sensor Integration and Hardware Design

The literature recently reports the effective combination of temperature, humidity, soil moisture, and pH sensors with low-cost microcontrollers to enable data acquisition [2], [4], [9].

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Capacitive soil moisture sensors have specifically been touted due to their dependability and decreased corrosion resistance [9]. Arduino and ESP32 platforms are flexible in terms of connecting a variety of sensors and communication modules, which enables scalability and deployment in the field [3], [4], [10]. Also, some of the designs include LCD or OLED screen and wireless connection, which enhance accessibility and real-time advantages to the end-users [1], [6], [4]. There are also some works which suggest modular systems that can be extended by new sensors or new features [10], [11].

C. Localized Advisory and Communication Systems

Systems are no longer limited to displaying uncooked sensor data, but can provide interpreted recommendations, frequently based on local agronomic recommendations and in the local language [7], [8]. Advanced analytics can be brought to the farmers in real time, even without the technical aptitude: the introduction of chatbots or messaging apps will provide the means of communication and remote consultation services [4], [7].

D. Limitations of previous works

Current systems of soil monitoring and advisory demonstrate the following limitations. Most of the solutions that have been implemented in the past have depended on direct NPK sensing technology, which is mostly costly and hence cannot be used by a large number of smallholders farmers. The advisory functions of previous systems are usually in the form of generic advice and do not accommodate localized advice in their native tongue, decreasing access and practical utility to the communities in the rural areas. The majority of systems are oriented to primitive data collection and visualization and little or no coordination of machine learning models able to forecast soil nutrients or produce adaptive and data-driven recommendations. A comparison of the previous works shown in Table I.

TABLE I: Comparison of Previous Works

Feature	[1]	[2]	[6]	[7]	[12]	[8]
ML-based NPK Prediction	X	X	X	X	✓	✓
Bangla Language Advisory	X	X	X	✓	X	X
Telegram Bot Remote Access	X	X	X	X	X	X
Local Policy Guidance	X	X	X	X	X	X
Low-Cost Modular Hardware	✓	✓	✓	✓	✓	✓

[a] Legend: ✓ = Feature Present, X = Not Present

III. Methodology

This project involved the complete design, implementation, and validation of an IoT-based soil health monitoring system SoilSense tailored for smallholder farmers. The overall workflow of the SoilSense system is illustrated in Fig. 1 and the stepwise operation of the SoilSense process is described in the flowchart in Fig. 5.

The work was done in the following major phases:

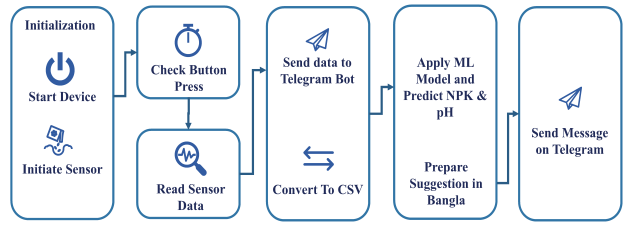

Fig. 1: Work flow of SoilSense system

TABLE II: Soilsense Hardware Components

Types	Component	Description	Functionality
Micro-controller	ESP32	Wi-Fi/Bluetooth microcontroller	Central control and wireless data transfer
Sensors	Soil Moisture Sensor	Measures soil moisture	Soil water monitoring for irrigation
	DHT11	Measures temperature and humidity	Tracks field climate for predictions
	Rain Sensor	Detects rainfall	Monitors rain for water management
Input/ Display	I2C 16 × 2 LCD	Text display with I2C	Shows readings/status to user
	Push Button	Tactile user switch	Triggers single readings
Power/ Connectivity	Power Supply	Provides system power	Field/lab power source
	Breadboard/ wires	Prototyping tools	Easy connections

A. Hardware Design and System Integration

Details of the hardware components used in SoilSense are listed in Table II. A block diagram showing the interconnection of all system components is provided in Fig. 2. The initial simulation setup is depicted in Fig. 3. The actual hardware implementation of the SoilSense system is shown in Fig. 4. The specific pin connections for each component are detailed in Table III.

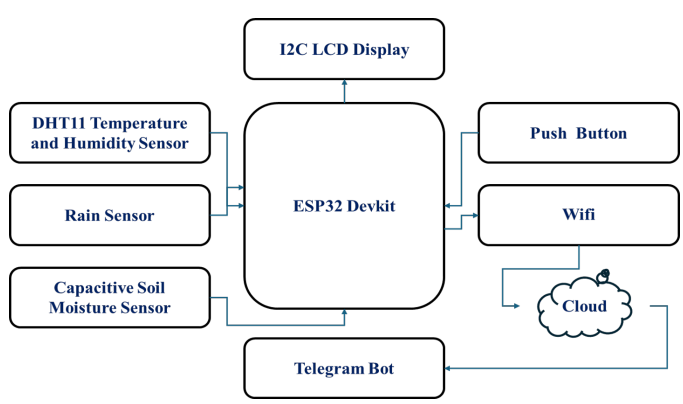

Fig. 2: Block diagram of the Soilsense system

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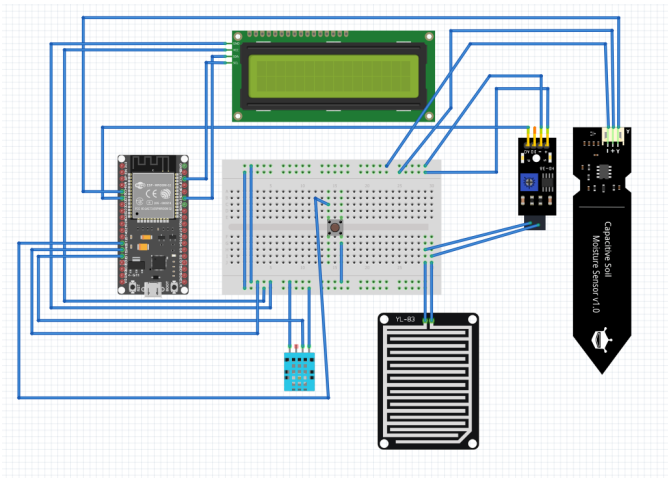

Fig. 3: Simulation Setup of SoilSense system

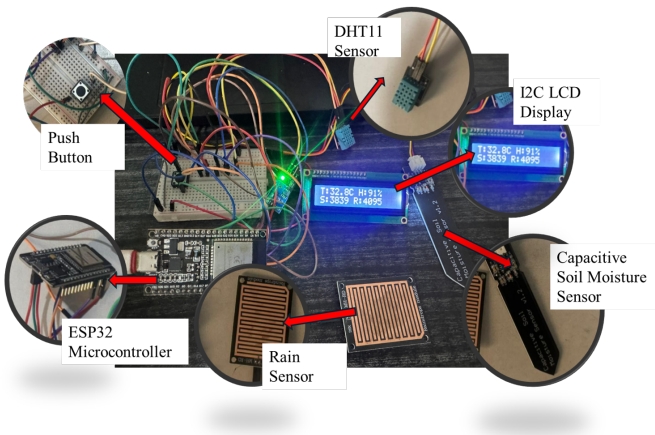

Fig. 4: Hardware Setup of SoilSense system

B. Data Acquisition and Communication

The ESP32 application code regularly (at a configured rate) reads all of the sensor channels. Data can be measured and shown immediately when a single-shot mode is used; when the long-term mode is set data is automatically stored with a 5-second interval during 10 minutes. A data record contains a timestamp and most recent values of the temperature, humidity, rainfall, and soil moisture. The system will show the current readings on the local screen and will send them through Wi-Fi to a Telegram bot through secure HTTP requests. To monitor over a long period, the ESP32 will output data tables in CSV format, which will be transmitted to the remote server to proceed with further processing. Farmers can access historical soil condition and prediction logs, which are suitable for agronomic planning, compliance with agricultural standards, or further research.

C. Machine Learning Model and Evaluation

A **Random Forest Regressor** was used to predict **N**, **P**, and **K** from the features **temperature**, **humidity**, and **rainfall**. The dataset [14] was split into **80%** training and **20%** testing. The Random Forest model works by constructing multiple decision trees, each trained on a random subset of the data. The final prediction is made by averaging the predictions from all trees in the forest. The prediction function for the Random Forest model is given by:

$$\hat{y}(X) = \frac{1}{N}\sum_{i=1}^{N} f_i(X) \tag{1}$$

Where: $f_i(X)$ is the prediction from the i-th tree, N is the total number of trees in the forest, X is the input features. The hyperparameters were used for the Random Forest Regressor shown in Table IV.

TABLE III: Soilsense Hardware Connection [13]

Component	Microcontroller Pin	Connection Details
DHT11 Sensor	GPIO 13 (DHT-PIN)	Data pin connected to GPIO 13; uses digital communication
Soil Moisture Sensor	GPIO 34 (SOIL_PIN)	Analog output connected to GPIO 34 (ADC input)
Rain Sensor	GPIO 35 (RAIN_PIN)	Analog output connected to GPIO 35 (ADC input)
Push Button	GPIO 12 (BUTTON_PIN)	Connected between GPIO 12 and GND; uses INPUT_PULLUP mode
LCD Display (I2C)	GPIO 21 (SDA), GPIO 22 (SCL)	Connected via I2C bus (SDA to GPIO 21, SCL to GPIO 22)
Power Supply	5V	Powered via 5V and ESP32 GND pins
Wi-Fi Network	N/A	ESP32 connects wirelessly to WiFi SSID and Password

TABLE IV: Random Forest Regressor Hyperparameters

Hyperparameter	Value
Number of Trees (`n_estimators`)	100
Max Depth (`max_depth`)	None
Min Samples Split (`min_samples_split`)	2
Min Samples Leaf (`min_samples_leaf`)	1
Random State (`random_state`)	42

The model was evaluated using the following metrics:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{2}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{3}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{5}$$

$$\text{Explained Variance} = 1 - \frac{\text{Var}(y - \hat{y})}{\text{Var}(y)} \tag{6}$$

Where: y_i is the actual value, $\hat{y}_i$ is the predicted value, $\bar{y}$ is the mean of the actual values, $\text{Var}(y)$ and $\text{Var}(y - \hat{y})$ are the variances of the actual values and residuals, respectively.

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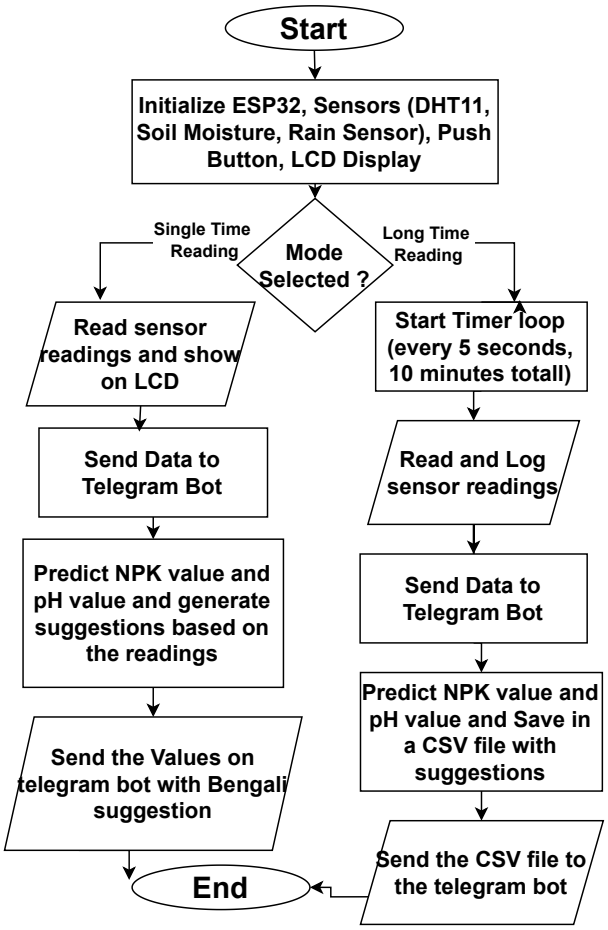

Fig. 5: Flowchart of SoilSense system

D. Real-Time Advisory and User Interface

Predicted soil nutrient values are automatically evaluated against threshold criteria established in the official Union Sahayika agricultural guidelines [15]. The system generates context-specific soil management advice, such as fertilizer recommendations. All advisory messages are delivered to the farmer in Bengali through the Telegram bot, ensuring accessibility and practical value. The Telegram interface supports both instant and batch queries, enabling farmers to receive on-demand insights and historical reports as needed. The mapping of predicted soil parameters to Bengali advisory messages is given in Table V.

IV. Results and Discussion

A. Data Collection and Graphical Analysis

The system collected real-time sensor data from three distinct soil types: Loamy, Sandy, and Clay, over a continuous period of 10 minutes at 5-second intervals. The sensor readings for temperature, humidity, rainfall, and soil moisture were used as inputs to predict NPK values using the machine learning model. The time-series variations of predicted N, P, K for loamy soil are illustrated in Fig. 6. Similar results for sandy soil are shown in Fig. 7, and for clay soil in Fig. 8. The visual analyses underscore differences in nutrient retention across soil types. The combination of automated data collection and graphical analysis enhances the understanding of soil health dynamics and supports better agricultural decision-making. Examples of Telegram messages for both single and long-term tests are shown in Fig. 9.

TABLE V: Recommendations based on Soil Parameters [15]

Parameter	Value Range	Recommendation (English Translated)
Nitrogen (N)	Very Low: < 50 kg/ha	Apply Urea
	Low: $50 - 90$ kg/ha	Apply Urea
	Medium: $90 - 130$ kg/ha	Apply Urea in lower quantity if necessary
	High: $130 - 170$ kg/ha	Urea is not needed
	Very High: > 170 kg/ha	Urea is not needed
Phosphorus (P)	Very Low: < 20 kg/ha	Apply TSP
	Low: $20 - 40$ kg/ha	Apply TSP
	Medium: $40 - 60$ kg/ha	Apply TSP in lower quantity if necessary
	High: $60 - 80$ kg/ha	TSP is not needed
	Very High: > 80 kg/ha	TSP is not needed
Potassium (K)	Very Low: < 40 kg/ha	Apply MOP
	Low: $40 - 80$ kg/ha	Apply MOP
	Medium: $80 - 120$ kg/ha	Apply MOP in lower quantity if necessary
	High: $120 - 160$ kg/ha	MOP is not needed
	Very High: > 160 kg/ha	MOP is not needed
Soil Moisture	Dry: ≤ 1000 (ADC Value)	Provide irrigation
	Optimal: $1000 - 3000$	Soil moisture is sufficient
	Excess: > 3000	There is excess water, stop irrigation

B. Machine Learning Model

The Table VI presents the evaluation metrics for each of the predicted items.

TABLE VI: Evaluation Metrics for Predictions of NPK

Metric	N	P	K
Mean Squared Error (MSE)	235.86	282.49	457.89
Root Mean Squared Error (RMSE)	15.36	16.81	21.40
Mean Absolute Error (MAE)	12.02	10.70	8.52
R-squared (R^2)	0.81	0.72	0.81
Explained Variance	0.81	0.73	0.81

C. Cost Analysis

Total estimated cost: 1750 BDT. A detailed breakdown of the total system cost is provided in Table VII.

TABLE VII: Cost Analysis in BDT

Category	Component	Estimated Cost (BDT)
Microcontroller	ESP32 Development Board	660
Sensors	Soil Moisture Sensor	280
	DHT11 Temperature & Humidity Sensor	150
	Rain Sensor	80
Display	I2C LCD 16x2	340
User Interface	Push Button	10
Power Supply	USB Power Cable / Adapter	80
Miscellaneous	Breadboard, Wires	150

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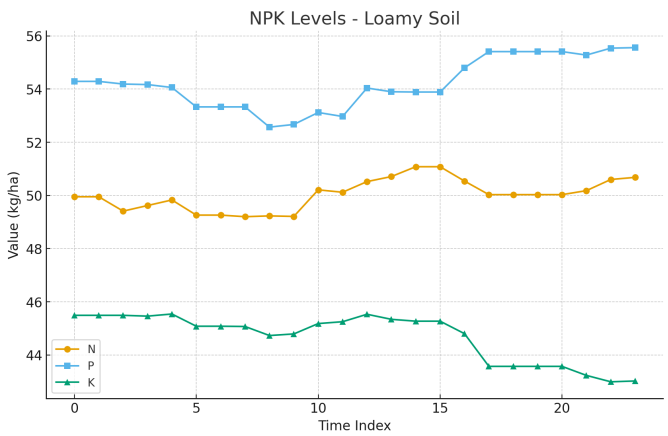

Fig. 6: Time-series Variation of Predicted NPK in Loamy Soil

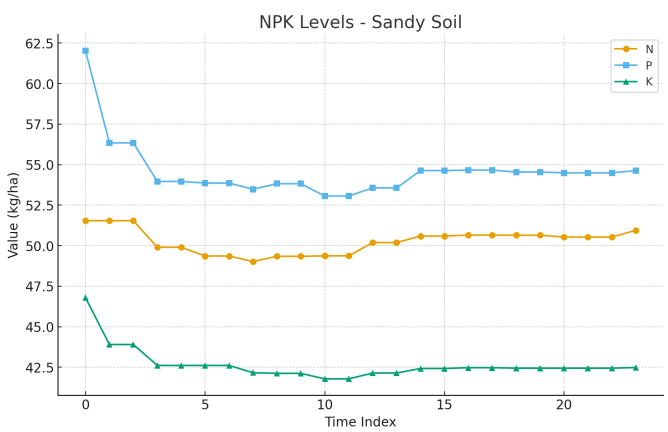

Fig. 7: Time-series Variation of Predicted NPK in Sandy Soil

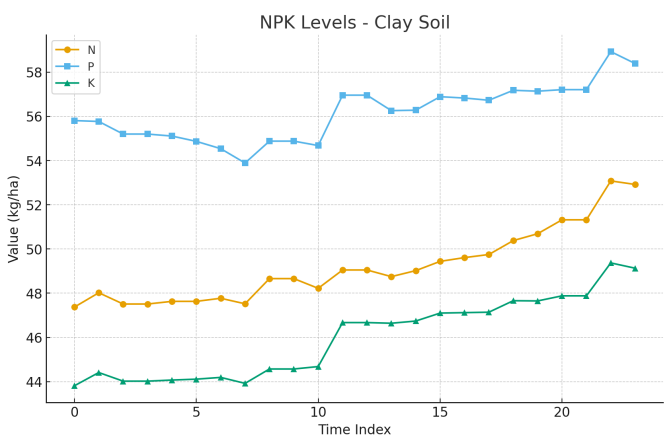

Fig. 8: Time-series Variation of Predicted NPK in Clay Soil

D. Discussion

The unified SoilSense system exhibits good functionality in the modules of sensing, communication, and prediction. The environmental sensors were able to give reliable real-time readings, which helped in the short and long term data collection. The Telegram interface allowed a smooth transmission of data and interaction with the user, CSV export, and delivery of advisory in local language, thereby making the system usable by smallholder farmers. The machine learning model generated predictive NPK with acceptable error values, demonstrating its aptitude in lightweight, IOT-based nutrient estimation. In general, the combination of these aspects of hardware sensing, wireless communication, and predictive analytics creates a solid decision-support workflow. As shown in Table VIII, shows that SoilSense does bring significant improvements with its ML-based nutrient forecasting, Bengali advisory support, and low cost modular architecture.

TABLE VIII: Comparison of Key Features with Related Works

Features	SoilSense	[1]	[3]	[6]	[10]	[7]
ML Integration	✓	X	X	X	X	X
Real-time Suggestion	✓	X	✓	✓	X	✓
Localized Language	✓	X	X	X	X	✓
Cost Efficiency	✓	X	✓	✓	✓	✓
Sensor Variety (NPK, pH, etc.)	✓	✓	✓	✓	✓	✓

[b] Legend: ✓ = Feature Present, X = Feature Not Present

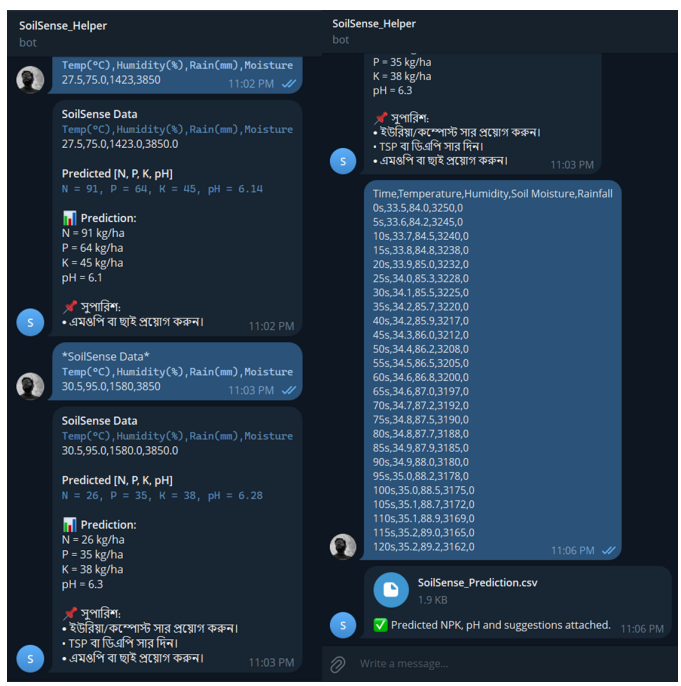

Fig. 9: Telegram Message Screenshot

E. Limitations and Recommendation

Despite promising outcomes, limitations are followings:

- **Dataset Precision:** The model predictions depend heavily on the accuracy and representativeness of the training

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dataset. More comprehensive, local-specific datasets will enhance the accuracy of predictions.

- **Guidebook Precision:** Crop-specific advisory recommendations will improve the actionable relevance of suggestions provided to farmers.
- **Updated ML Models:** Apart from Random Forest Regression, there are more optimal and scalable models which can be taken in consideration.

Addressing these limitations will significantly strengthen future implementations of the SoilSense system.

V. Conclusion

SoilSense is an IoT-based soil health monitoring and advisory system that can assist smallholder farmers in Bangladesh with real-time soil measurements, NPK forecasts, and localized Bengali advice using a single Telegram bot, which has been tested on loamy, clay and sandy soils. Although the system is efficient at integrating machine learning, low-cost hardware, and user-friendly communication, there are still issues of accuracy of datasets, sensor calibration, and the overall advisory accuracy. The improvement of such areas by means of better data collection, cooperation with research institutes, better calibration, and comparing with NPK and pH sensors will make the system more reliable. Among the future merits are crop-specific and location-specific suggestions, irrigation and pest management directions, and additional means of communication, including SMS and voice support, which would guarantee the spread of sustainability in agricultural practices and extend its influence.

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